

ViralForge AI

Project Specification — The Autonomous AI Creative Director

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1. Overview

ViralForge AI is an autonomous AI creative director for short-form social video. Given a single topic (optionally steered by a creator's interests or social profiles), it runs a ten-step pipeline with no human handoffs: it discovers rising stories on the live web, builds a trend knowledge graph, chooses a creative strategy, writes hooks, generates all media (image, video, voice, music, captions), produces three complete video variants, scores them A/B/C, learns from the losers, assembles the winner with VEED, and delivers a publish-ready 9:16 vertical video.

The entire run is visualized in a real-time **3D mission-control interface** (React Three Fiber): the knowledge graph floats in space, media jobs fan out from a central core, variants play in a spatial theatre, and the camera choreographs itself to each pipeline stage.

10 steps → 1 click

3 variants per run (A/B/C)

Benchmark 0.71 → 0.89 (frontier 0.91)

Publish-ready 9:16 output

2. Problem Statement

- **Trends decay in hours.** By the time a creator spots a rising story, briefs an editor, and ships, the spike is gone.
- **Fragmented tooling.** Research, scripting, image, video, voiceover, music, captions, and editing each live in a different product.
- **Creative guesswork.** Hook, story, and CTA choices are gut feel; learning happens only after a post flops.
- **No learning loop.** Every video is produced with the same blind process as the first.

3. Goals & Non-Goals

Goals

- Fully autonomous trend-to-video pipeline triggered by one topic prompt.
- Quality *prediction and improvement* via the Pioneer Social Video Intelligence Model, fine-tuned on synthetic GOOD vs BAD creative pairs.

- fal.ai as the single media operating system — all image, video, and audio generation plus VEED Subtitles and Fabric 1.0 through one API key.
- A cinematic 3D command-center UI that makes the autonomous process legible and demo-ready.
- Graceful degradation: the demo narrative always completes, even when an upstream API fails.

Non-Goals (hackathon scope)

- No persistence layer — pure demo, no database or accounts.
- No direct social-platform publishing (output is a publish-ready file).
- No multi-user or workspace features.

4. Pipeline Specification

#	Stage	Service / Model	Behavior
1	Discover	Tavily Search API	Searches the live web for rising stories on the topic, steered by creator interests/social profiles. Parses social handles (e.g. @helshahaby) from the topic into the interest field automatically.
2	Understand	GLiNER2 (fastino/gliner2-large-v2 , HF Inference) + Pioneer	Extracts entities from research; Pioneer organizes them into a Trend Knowledge Graph rendered as an interactive 3D node graph.
3	Direct	Pioneer (google/gemini-3.7-flash via Lovable AI Gateway)	Selects the strongest story angle and issues a creative verdict. Prompts enforce strict topic relevance — every scene must concretely depict the subject (e.g. humanoid robots in clinics).
4	Hook	Pioneer	Writes three competing hooks for variants A, B, C.
5	Generate media	fal queue API	Fan-out: hero image (fal-ai/z-image/turbo), 3 B-roll clips (fal-ai/flux-3/text-to-video/draft , audio-enabled, up to 20s), voiceover (fal-ai/kokoro/american-english), music (CassetteAI/music-generator), captions. Async submit/poll with per-job status.
6	Multiply	—	Composes three full video variants (A/B/C), played simultaneously in a floating 3D theatre.
7	Evaluate	Pioneer judge	Scores each variant on hook, story, visuals, and CTA; explains the verdict; records failures as learning signal for future runs.
8	Assemble	VEED via fal: veed/subtitles , veed/fabric-1.0	Submits the winning cut for captioning (hustle preset) and a Fabric 1.0 pass; polls jobs and surfaces final MP4 links. Authentication and billing go through the fal key — no standalone VEED key.
9	Deliver	—	Presents the publish-ready 9:16 video with sound toggle; browser autoplay handled via one-time user-gesture unmute.

5. Pioneer — Social Video Intelligence Model

- **Purpose:** predict and improve the quality of social-video creative decisions (hook, story, CTA, scene direction).

- **Synthetic Fine-Tune Lab:** an in-app lab that generates thousands of synthetic GOOD vs BAD training pairs per creative dimension and simulates the fine-tune run.
- **Benchmark demonstration:** general model **0.71** → fine-tuned Pioneer **0.89** → frontier reference **0.91**, shown live in the HUD.
- **Learning loop:** judge verdicts and losing variants feed back as preference signal for subsequent runs.

6. Architecture

Frontend

- TanStack Start v1 (React 19, SSR) + Vite; Tailwind CSS v4 with semantic design tokens (dark mission-control theme).
- React Three Fiber + drei scene: `KnowledgeGraph3D`, `VariantFrames`, `CameraRig` with per-stage camera choreography.
- Zustand store (`src/lib/pipeline/store.ts`) tracks phases, media, scores; the orchestrator (`orchestrator.ts`) drives the 10-step sequence.
- HUD: top bar with HUD visibility toggle, log stream, stage panel (scores, VEED job status, final player with sound control), synthetic lab.

Backend (server functions)

- `tavily.functions.ts` — trend discovery. `gliner.server.ts` / `extract.functions.ts` — entity extraction.
- `pioneer.server.ts` / `pioneer-core.server.ts` — story, hooks, scripts, judging, synthetic lab (8k max output tokens for long scripts).
- `fal.server.ts` / `fal.functions.ts` — generic `falSubmitRaw` + poll against fal queue endpoints; model registry for image/video/voice/music.
- `veed.server.ts` / `veed.functions.ts` — VEED Subtitles + Fabric 1.0 jobs submitted through fal.

Security

- All provider keys (`FAL_KEY`, `TAVILY_API_KEY`, `HF_TOKEN`, `PIONEER_API_KEY`, `H_API_KEY`, `LOVABLE_API_KEY`) are server-side secrets, read inside handlers only; never shipped to the client bundle.

7. Resilience & Fallbacks

Every live stage has an on-topic bundled fallback (health-training × humanoid-robotics theme): hero image, three 10s clips, a 24.5s neural voiceover, a music bed, and a pre-composed ~31s narrated final cut. fal API calls require account balance — the dashboard's daily free generations apply to the web playground, not the API. The UI surfaces real statuses (e.g. fal 403 exhausted balance) in the log stream while the demo completes via fallbacks.

8. Deliverables

- Live 3D demo application (this project).
- Pitch deck: `ViralForge-AI-Hackathon-Deck.pptx` (5 slides).
- This specification (PDF) and project `README.md` .
- 2-minute narrated demo video: `ViralForge-AI-Demo.mp4` .

9. Future Work

- Persist runs and learning signal in Lovable Cloud; cross-run memory for the judge.
- Real fine-tune export of synthetic pairs to Pioneer's training API.
- Direct publish integrations (TikTok / Reels / Shorts) and analytics feedback loop.
- H Company agent-driven VEED editor session for hands-free final assembly.